Worked example: mapping raw value $r = 500$ to bin $s = 10$
($min = 350$, $max = 650$, $b = 20$ bins)

Step 1 — compute range: $\delta = (max + 1) - min = (650 + 1) - 350 = 301$

Step 2 — bin capacity: $cap = \delta\,/\,b = 301\,/\,20 = 15.05$

Step 3 — scaled index: $s = \mathrm{int}\left(\dfrac{r - min}{cap} + 1\right) = \mathrm{int}\left(\dfrac{500 - 350}{15.05} + 1\right) = \mathrm{int}(10.97) = 10$